
Mobile and Ubiquitous Computing

UC Presentation



FCUL, Departamento de Informática

Ano Letivo 2026/2027

Ana Paula Afonso

Diogo Branco

Staff and communication

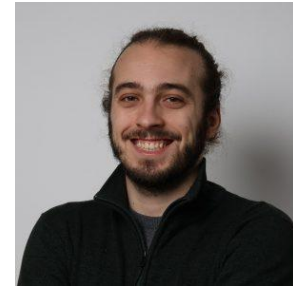
- Theoretical classes

- Ana Paula Afonso (coordinator)



- Laboratory classes

- Diogo Branco



- Communication

- docentes-cm@listas.di.ciencias.ulisboa.pt
- Web page: <https://moodle.ciencias.ulisboa.pt/course/view.php?id=6706>
News and Forum

...

Schedule 26/27

	Seg 9/21	Ter 9/22	Qua 9/23	Qui 9/24	Sex 9/25
07:00					
08:00					
09:00					
10:00					
11:00					
12:00					
13:00			13:00 - 15:00 T 1.3.15		
14:00					
15:00			15:00 - 16:30 TP 1.2.21		
16:00					
17:00			16:30 - 18:00 TP 1.2.21		
18:00					

Objectives

- Understand the software development for Ubiquitous Systems and Applications, considering

The Role of Context, Use, and Technology

The Challenges of Design, Development, and Evaluation Processes

The Frameworks for development and supporting OS

The Human Perspective

The Data Coherence and Storage Perspective

The Systems' Perspective

The Trends and Future Endeavors

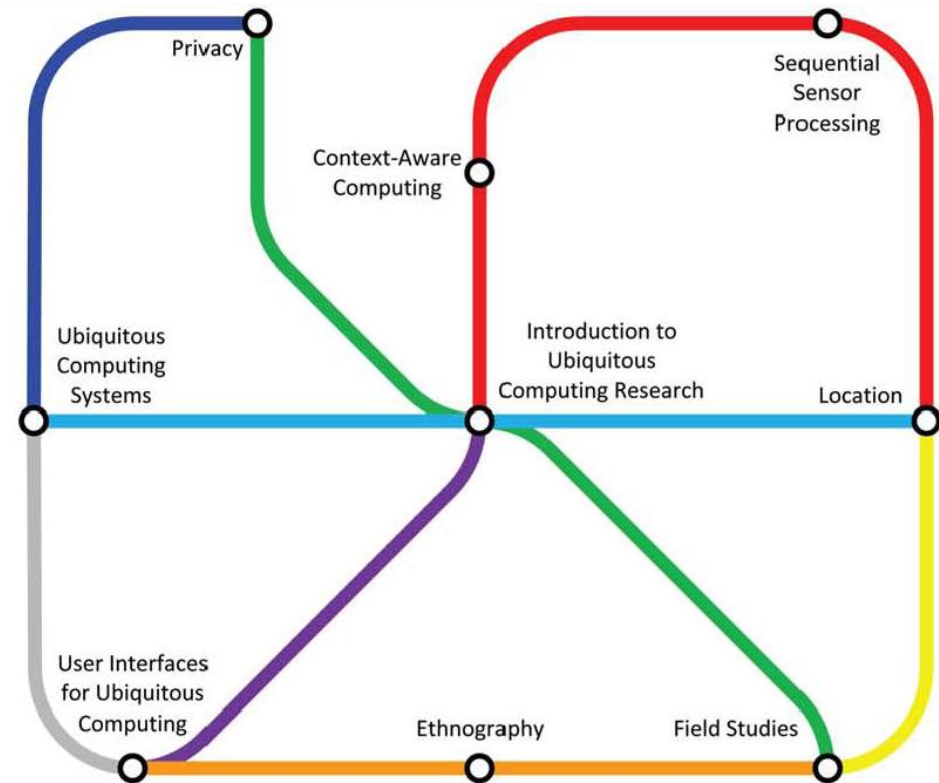


Image from John Krumm

Mobile & ubiquitous computing

Topics

- Mobile interaction design
- Mobile prototyping
- Mobile evaluation
- Wireless communications and networks
- Location and positioning systems
- Context-aware computing
- Sensors processing

Technologies

- Frameworks for mobile application design
- OS specifics
 - Google Android
 - Android Studio IDE
 - Kotlin (language recommended by Google since 2019)

Previous knowledge

- Recommendations for successfully engaging in this course include the following courses in a 1st-cycle Informatics degree (or equivalent):
 - Object-Oriented Design (Object-Oriented Programming)
 - Databases and Information Systems
 - Human-Computer Interaction
 - Distributed Systems
 - Computer Networks

Evaluation

- Components
 - **Theoretical assignment (30%)**
Topic analysis, presentation, and discussion in T classes + quiz (30%)
 - **Laboratory assignments (15%)**
within Lab classes
 - **Project (55%)**
Project Proposal Presentation + 2 Project checkpoints (15%)
Final deliverables & Open Studio (40%)

Theoretical assignment and project components must score at least 9.5 for course approval

Students with formal specific status should send a request for alternative evaluation to the course email (slide 17)

Theoretical Assignment

Topic analysis, presentation and quiz

- **Group assignment**

- **Individual evaluation**

- Discussion of research topics in T classes
 - Topics will be available on the course web page (October)
 - Each topic will be assigned with one research paper
 - An email should be sent with an ordered list of 6 topics for presentation
 - Presentation and evaluation: see CM plan
 - **Each group is assigned to 1 topic ...**

... and should:

- **Write a paper** (min 6pg – max 8pg, PDF) describing the topic and making use of relevant papers in that domain
- Each topic should be divided into 3 main sub-topics, one for each student
- Each student/sub-topic should discuss at least one paper (not considering the example paper), but students are encouraged to look for more papers to provide a more complete description of the topic.
- **Make a presentation** (15 min)
- Deliver the paper and presentation through the Moodle activity one day before the presentation date
- In the T class, be prepared to present, criticize, and contest the critique in class

Theoretical Assignment

Topic analysis, presentation and quiz

- The paper should:
 - Introduce the topic, summarizing challenges, opportunities and/or trends in its exploration
 - Analyze 3 sub-topics in light of prior work, being critical (positively and/or negatively) about the works presented
 - Discuss opportunities and future trends
 - Clearly indicate the contributions of each student to the paper, ensuring that each one is responsible for 1 sub-topic.
 - Clearly indicate if LLMs were (or not) used, and how they were used.
 - Comply with the ACM 2-Column format
- Quiz
 - Following the presentation, all students take an individual quiz on that topic, which also connects it to the fundamental content covered across all six first theoretical classes
- Grading
 - **Failing to send the documentation, present, and complete the quiz results in a null score**
 - No second chances
 - Individual student evaluation
 - Minimum score: 9.5

Theoretical Assignment

LLM Usage Policy – What we expect

- Reading and summarizing papers: must be done without the use of LLMs. The critical analysis has to reflect the students' own work and understanding
- Mandatory disclosure: the paper must clearly state whether LLMs were used and, if so, how (e.g., grammar review, formatting, etc.)
- Not allowed: using LLMs to generate the summary, critical analysis, or conclusions about the sub-topics. This is individual work for each student on their assigned sub-topic
- Permitted only for supporting tasks (e.g., spelling/grammar correction, ACM formatting), provided it is properly disclosed
- Goal/expectation: ensure the critical analysis and understanding of the topic are genuine, demonstrating each student's own work on their sub-topic

Lab Assignments

- Individual assignments
 - Individual evaluation
- 5 Assignments
 - Training guides + exercises
- The process:
 - Training guides and samples
To be consumed in class
 - The exercises
Launched in a class and evaluated in the next one
- Evaluation
 - The implementation, demonstration, and knowledge contribute to the grade (demo + code)
- Grading
 - Failing to demonstrate corresponds to a zero grade
 - No second chances
 - The 5 assignments correspond to **15%** of the course classification

Project

- Objective
 - Build an App
- Basic requirements
 - Good user experience
 - Feature rich (service integration, ..)
 - Good data management
 - Distributed
 - Adequate combination of the above or others
- More requirements
 - Creativity, arguments, design planning
 - Bibliography, writing, and structure
 - Fail to mention a bibliography could be considered fraud
- Themes (in next week's labs)
- Group size
 - **3 students**
 - **Group enrollment on Moodle**
- Available mobile equipment & lab software conditions

Project

Deliverables

Project Proposal Presentation

- It should motivate the concept, present related work, and describe the approach and features
- Submission: one day before the presentation
- Presentation and Discussion: see CM Plan

Project final deliverables

- Project Report
 - 1 short paper (2 pages)
- Source code & App apk
- Promotional video (less than 2 min)

Checkpoints

- App presentation & demo
- Dates: see CM plan

Open Studio/demo

- Final Project demo
- “Oral exam” with teachers & external jury
- **20 January**

Evaluation

- Project proposal presentation + 2 checkpoints (**15%**)
- Project final deliverable & Open Studio (**40%**)
- Minimum score: 9.5

Project: Open Studio demo videos

Videos @ MUC web page from past editions

Planning – Theoretical, T/P classes (tentative)

Semana		Theoretical Classes			Laboratory Classes
Nº	Date	Week	Date	Summary	Summary
1	14-Sep	1	16/set	Course presentation: program, assignments and evaluation	-
	15-Sep				
	16-Sep				
	17-Sep				
	18-Sep				
2	21-Sep	2	23/set	Introduction to mobile and ubiquitous computing: understanding the problems	Introduction. Android Studio. Kotlin. Preparing first hands-on lab.
	22-Sep				
	23-Sep				
	24-Sep				
	25-Sep				
3	28-Sep	3	30/set	Designing and evaluating mobile applications	Android Basic GUI & Navigation. Lab assign evaluation #1 - First App
	29-Sep				
	30-Sep				
	01-Oct				
	02-Oct				
4	05-Oct	4	7/out	Location in ubiquitous systems: location technologies; location systems	App Architecture & Getting data from the Web. Lab assign evaluation #2 Multiple Screens App
	06-Oct				
	07-Oct				
	08-Oct				
	09-Oct				
5	12-Oct	5	14/out	Wireless communication and wireless networks	Persistant Data. Lab assign #3 - API use, Data Feeth & Display
	13-Oct				
	14-Oct				
	15-Oct				
	16-Oct				
6	19-Oct	6	21/out	Context-aware computing: classification, problems, and sensors	Android Sensors. Lab assign #4 - Remote Local Storage
	20-Oct				
	21-Oct				
	22-Oct				
	23-Oct				
7	26-Oct	7	28/out	No classes (Ciências Research Day)	No classes (Ciências Research Day)
	27-Oct				
	28-Oct				
	29-Oct				
	30-Oct				

Planning – Theoretical, T/P classes (tentative)

Semana		Theoretical Classes			Laboratory Classes	
Nº	Date	Week	Date	Summary	Summary	
8	02-Nov	8	4/nov	Themes' discussion: session	Project Released. Lab assign #5 - Using Sensors	
	03-Nov					
	04-Nov					
	05-Nov					
	06-Nov					
9	09-Nov	9	11/nov	Themes' discussion: session	Project Proposal	
	10-Nov					
	11-Nov					
	12-Nov					
	13-Nov					
10	16-Nov	10	18/nov	Themes' discussion: session	Project - Low fidelity prototyping	
	17-Nov					
	18-Nov					
	19-Nov					
	20-Nov					
11	23-Nov	11	25/nov	Themes' discussion: session	Checkpoint 1 - Project GUI	
	24-Nov					
	25-Nov					
	26-Nov					
	27-Nov					
12	30-Nov	12	2/dec	Themes' discussion: session	Project - sensors, db	
	01-Dec					
	02-Dec					
	03-Dec					
	04-Dec					
13	07-Dec	13	9/dec	Themes' discussion: session	Project - multi-user, services	
	08-Dec					
	09-Dec					
	10-Dec					
	11-Dec					
14	14-Dec	14	16/dec	Themes' discussion: session	Checkpoint 2 - Alpha version	
	15-Dec					
	16-Dec					
	17-Dec					
	18-Dec					
			20/jan	Open Studio: Final Project Demo & Deliverables		

Evaluation of working students

- Evaluation is similar to regular students, except:
 - Lab Assignments

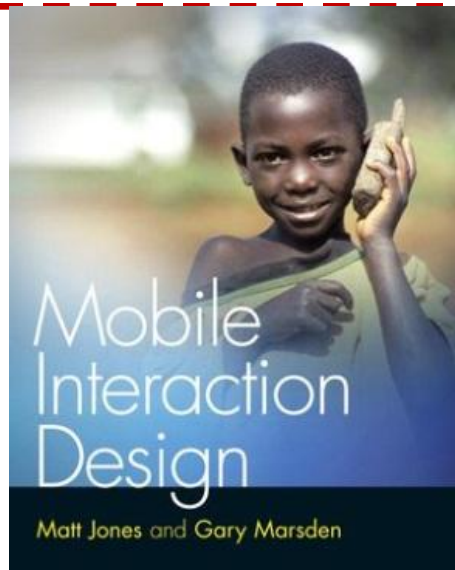
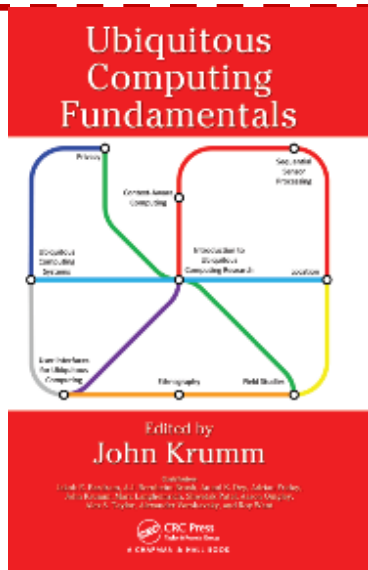
A challenge is posted weekly

An email should be sent by next Wednesday at 23:59, with the link to the solution on GitHub and the APK that accomplishes the objectives
 - Project checkpoints

Presentation/demo of checkpoints can be pre-recorded (video) and shared before class

Theoretical assignment and the final project demo studio are equal

Bibliography



Ubiquitous Computing Fundamentals. John Krumm. CRC Press

Mobile Interaction Design. Matt Jones, Gary Marsden. Wiley

Mobile Design and Development: Practical concepts and techniques for creating mobile sites and web apps. Brian Fling. O'Reilly

More book references on the webpage

Papers available on the webpage

QUESTIONS?